

Hawai'i Hurricane Lala: Facebook Data Situation Report

NSF AI-SDM and Meta AI for Good Draft hurricane situation report

Meta Disaster Analysts

Disaster Manager Brief - Hurricane Lala, Hawai'i

Key Takeaways

- **Data period.** The population series spans **August 15, 2026 12am through August 29, 2026 4pm**, with one observation every eight hours. The statewide storm-path context figure is shown at **August 16, 2026 8am**.
- **Population change.** Red tiles indicate fewer Facebook users than the historical baseline; green tiles indicate more users. The maps use a symmetric **dark red → white → dark green** scale.
- **Terrain context.** Elevation is converted to a raster and then to contour lines using an `elevatr + terra::as.contour()` technique. Major index contours are emphasized to make Hawai'i Island's topography easier to interpret.
- **Storm track.** A solid black line represents the realized NHC/CPHC best track through the time of each map. The official path predicted at or before that map time is shown separately as a **red dashed line**.
- **Map hierarchy.** Shaded terrain, rivers, and hurricane-track information are drawn below the population surface. The Bing population polygons remain visually dominant, while **major roads are drawn on top of the Bing tiles** so the transportation network stays easy to read. City labels and the current storm-center marker remain above the roads.
- **City selections.** Population time series use exact **12 × 12 mile** square selections centered on Hilo, Kailua-Kona, Waikoloa Village, and Pāhoā.

Table 1: Cities used in the Big Island analysis and the population change in each 12 × 12 mile selection during the first available window.

Place	Role	Latitude	Longitude	Net change in users	Tiles with data
Hilo	East Hawai'i population hub	19.7074	-155.0885	288	42
Kailua-Kona	West Hawai'i population hub	19.6400	-155.9969	231	35
Waikoloa Village	Northwest Hawai'i community	19.9400	-155.7900	-4	21
Pāhoā	Puna district community	19.4975	-154.9517	399	45

Overview

This report describes population movement on Hawai'i Island during Hurricane Lala using Meta's *Facebook Population During Crisis* data from August 15, 2026 12am through August 29, 2026 4pm. Population observations occur every eight hours.

The detailed Big Island maps combine the population signal with terrain contours, major rivers, major roads, and NHC/CPHC storm-track information. The Bing population layer is intentionally near the top of the visual hierarchy so population changes remain visually dominant, while major roads are drawn above the tiles to keep the transportation network legible.

1. Statewide Hawai'i storm-path context on August 16

This statewide view is shown at **August 16, 2026 8am** so the realized approach of the storm and the official forecast available at that point can be interpreted in relation to the Hawaiian Islands as a whole.

August 16, 2026 8am

As Hurricane Lala Downgraded to Tropical Storm | solid black = realized path; red dashed = official forecast

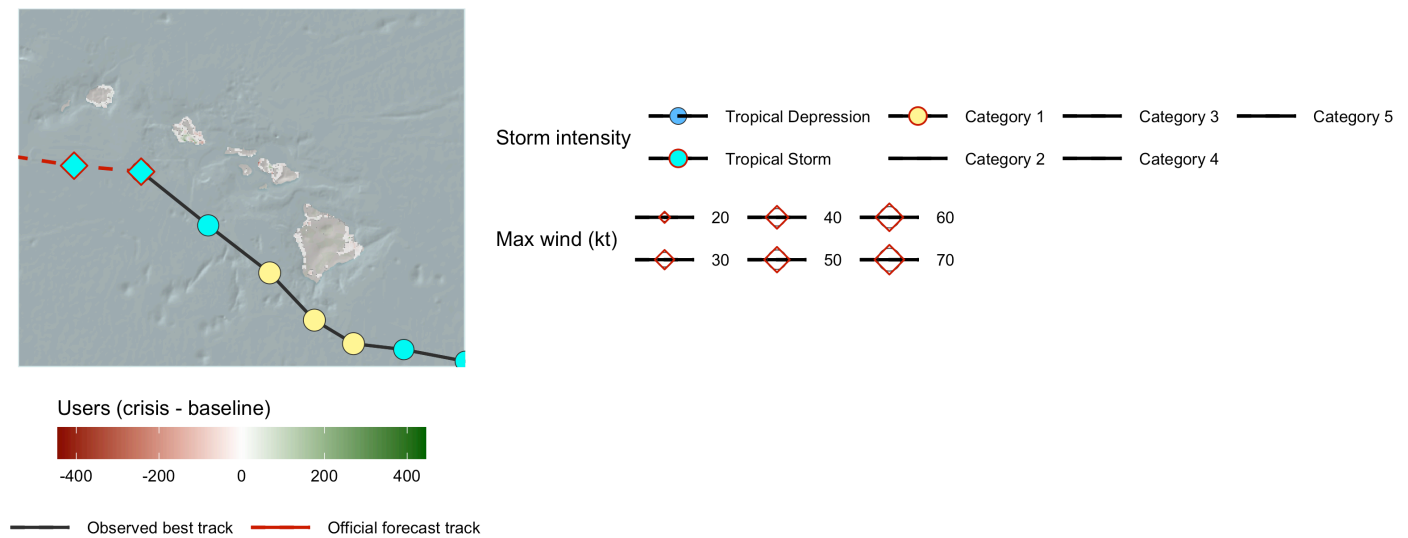


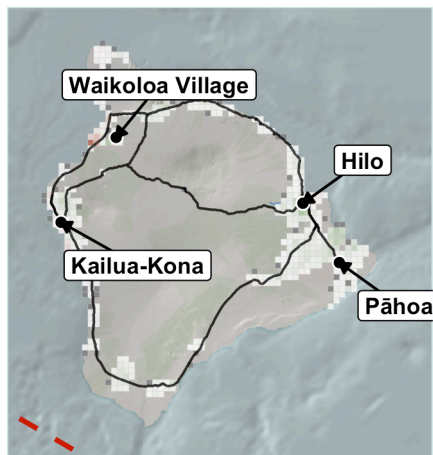
Figure 1: Statewide Hawai'i view at August 16, 2026 08:00 HST. The observed track is shown with circular intensity markers; the official forecast uses a red dashed path and diamond markers.

2. Big Island population change: Evolution through the crisis

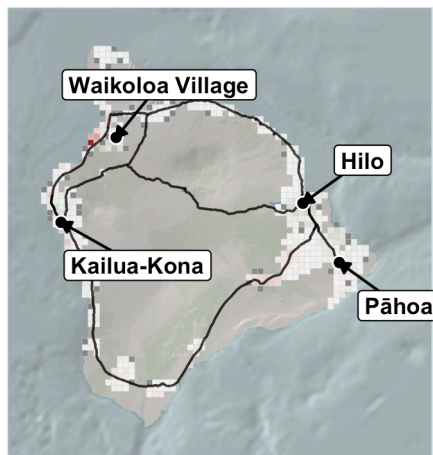
Dark red = fewer users than baseline; dark green = more users than baseline; white = little/no net change; grey = privacy-threshold / missing data.

The detailed maps use terrain-contour method and make major roads more visible. The Bing population cells sit second-to-top so the population pattern is the dominant visual element. The three panels below correspond exactly to the start, midpoint, and end of the Hurricane Lala data collection.

August 15, 2026 12am



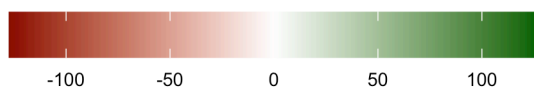
August 22, 2026 8am



August 29, 2026 4pm



Users (crisis - baseline)



colour ———— Observed best track

Figure 2: Population difference at the start, exact midpoint, and end of the event period. All panels use the same population-change scale.

Population change in selected 12 × 12 mile city regions



Figure 3: Net change in users (crisis – baseline) within exact 12 × 12 mile square selections centered on each Big Island city, in 8-hour intervals.

Appendix

Data description

The analysis uses Meta's **Facebook Population During Crisis** data. The signal represents Facebook app users with Location Services enabled, not the full resident population. For this event the population series runs from **August 15, 2026 at 12:00 am HST through August 29, 2026 at 4:00 pm HST**, in **8-hour increments**.

Population measurements are provided in Bing level-14 tiles. Crisis counts are compared with the historical baseline for the corresponding time period.

For city-level summaries, each city is centered within an exact **12 × 12 mile square** constructed in a local metric projection before population tiles are spatially selected.

Map, terrain, and storm data

The terrain implementation intentionally follows the Nepal flood-map technique: elevation is downloaded through `elevatr`, projected with `terra`, converted into contour lines using `terra::as.contour()`, and cached locally. Major contour lines are emphasized to make the landform easier to read.

Major rivers and roads are queried from OpenStreetMap and cached locally. The code rotates across multiple Overpass endpoints if a server times out.

NHC/CPHC ATCF data provide the observed best track and official forecast track. For retrospective reporting, the hurricane helper checks both current ATCF aid files and the yearly ATCF archive. The forecast selected for a map is the latest official (0FCL) forecast initialized at or before that map's time.

The map distinguishes storm history from prediction:

- **Solid black:** realized / observed best track through the map time.
- **Red dashed:** official predicted center track available at or before the map time.
- **Blue translucent area:** forecast uncertainty cone when a specific archived cone KMZ is supplied.
- **Grey contours:** elevation.
- **Black road lines:** major road network, drawn above the Bing population layer.
- **Blue lines:** major rivers.
- **Dark red → white → dark green cells:** population change relative to baseline.

Data caveats

- **Sample bias.** The data cover Facebook users with Location Services enabled, not all residents or visitors.
- **Data intervals.** Population data are aggregated into 8-hour windows.
- **Privacy controls.** Suppressed / missing tiles should not be interpreted as zero population change.
- **Baseline comparison.** Crisis differences are relative to a historical baseline, not direct evacuation counts.
- **Connectivity.** Apparent population decreases can reflect evacuation, power loss, cell-service disruption, or a combination.
- **Terrain.** Hawai'i Island's topography can constrain travel and isolate communities; movement patterns should be interpreted alongside the contour lines and road network.
- **Forecast vs. observed track.** The forecast path represents what was predicted at that point in the event; the best track is the retrospective observed/analysed storm path.
- **Users ≠ people.** Facebook-user counts are not direct counts of residents, evacuees, vehicles, or shelter demand.

Sources

- Meta: Facebook Population During Crisis.
- NOAA National Hurricane Center / Central Pacific Hurricane Center: ATCF best-track and official forecast products.
- Map data © OpenStreetMap contributors.
- Terrain from AWS Terrain Tiles through the `elevatr` package.
- Basemap tiles through `maptiles`.